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Faculty of Computer Science

Knowledge Management & Discovery Lab

Exercise 9

1. What is the difference between internal and external cluster validation indices?

Given are 15 instances $\{x_1, \dots, x_{15}\}$ and three classes $\{A, B, C\}$. The class assignment is as follows:

- $A = \{x_1, x_2, x_3, x_4, x_5\}$
- $B = \{x_6, x_7, x_8, x_9, x_{10}\}$
- $C = \{x_{11}, x_{12}, x_{13}, x_{14}, x_{15}\}$

Executing K-Means with (a) $K = 3$ and (b) $K = 5$ results in the two imperfect clusterings Z_1 and Z_2 :

Z_1 :

- $X_1 = \{x_1, x_2, x_3\}$
- $X_2 = \{x_5, x_6, x_7, x_9, x_{10}, x_{15}\}$
- $X_3 = \{x_4, x_8, x_{11}, x_{12}, x_{13}, x_{14}\}$

Z_2 :

- $X_1 = \{x_1, x_2, x_3\}$
- $X_2 = \{x_5, x_6, x_7\}$
- $X_3 = \{x_9, x_{10}, x_{15}\}$
- $X_4 = \{x_8, x_{11}, x_{12}\}$
- $X_5 = \{x_4, x_{13}, x_{14}\}$

Calculate the purity for each of the clusterings and compare the results. What is the problem of this measure?

2. Given are 6 instances $\{p_1, \dots, p_6\}$ and three classes $\{A, B, C\}$. The class assignment is as follows:

- $A = \{p_1, p_2\}$
- $B = \{p_3, p_4\}$
- $C = \{p_5, p_6\}$

Executing K-Means with $K = 3$ results in the imperfect clustering Z_3 :

Z_3 :

- $Y_1 = \{p_1, p_2\}$
- $Y_2 = \{p_3, p_5\}$
- $Y_3 = \{p_4, p_6\}$

To compute the Rand Index and the Jaccard Coefficient, compare the class similarity matrix with the cluster similarity matrix. In both matrices, the diagonal entries are ignored. Count each unordered pair only once, for example by using only the upper triangular part of the matrices.

The class similarity matrix is defined as follows:

$$\text{ClassSim}(p_i, p_j) = \begin{cases} 1, & \text{if } p_i \text{ and } p_j \text{ belong to the same class,} \\ 0, & \text{otherwise.} \end{cases}$$

For the given class assignment, the class similarity matrix is:

	p_1	p_2	p_3	p_4	p_5	p_6
p_1	—	1	0	0	0	0
p_2	1	—	0	0	0	0
p_3	0	0	—	1	0	0
p_4	0	0	1	—	0	0
p_5	0	0	0	0	—	1
p_6	0	0	0	0	1	—

The cluster similarity matrix is defined as follows:

$$\text{ClusterSim}(p_i, p_j) = \begin{cases} 1, & \text{if } p_i \text{ and } p_j \text{ belong to the same cluster,} \\ 0, & \text{otherwise.} \end{cases}$$

For the clustering Z_3 , the cluster similarity matrix is:

	p_1	p_2	p_3	p_4	p_5	p_6
p_1	—	1	0	0	0	0
p_2	1	—	0	0	0	0
p_3	0	0	—	0	1	0
p_4	0	0	0	—	0	1
p_5	0	0	1	0	—	0
p_6	0	0	0	1	0	—

For each unordered pair (p_i, p_j) , compare the corresponding entry in the two matrices. The first subscript refers to the cluster similarity value and the second subscript refers to the class similarity value:

f_{11} : same cluster and same class.

f_{10} : same cluster but different class.

f_{01} : different cluster but same class.

f_{00} : different cluster and different class.

The Rand Index and Jaccard Coefficient are then computed as follows:

$$\text{RandIndex} = \frac{f_{11} + f_{00}}{f_{11} + f_{10} + f_{01} + f_{00}}, \quad \text{JaccardCoefficient} = \frac{f_{11}}{f_{11} + f_{10} + f_{01}}.$$

Compute the Rand Index and Jaccard Coefficient of clustering Z_3 .

3. **Calculate cohesion and separation for Y_1 in Z_3** assuming the data points have the coordinates depicted in Table 1 and Figure 1.

Use the centroid-based cohesion and separation formulas from the lecture slides. Since the clustering was generated by K-Means, use the cluster centroids as representatives of the clusters.

For this dataset, the largest Euclidean distance between two data points is between $p_2 = (2, 5)$ and $p_6 = (4, 1)$:

$$d_{\max} = d_E(p_2, p_6) = \sqrt{20}.$$

Use this value to normalize all distances needed for the cohesion and separation calculations:

$$d_{\text{Normalized}}(a, b) = \frac{d_E(a, b)}{d_{\max}} = \frac{d_E(a, b)}{\sqrt{20}}.$$

This normalization must also be applied to distances involving centroids (not for computing them), including the centroid-to-centroid distances used for separation.

First compute the centroids of the three clusters and then complete the following table for the distances of each point to the centroid of its own cluster:

Point	Cluster	$d_E(p, \text{center}(Y_i))$	$d_N(p, \text{center}(Y_i))$
p_1	Y_1		
p_2	Y_1		
p_3	Y_2		
p_5	Y_2		
p_4	Y_3		
p_6	Y_3		

Finally, compute the normalized distances between the cluster centroids:

d_N	center(Y_1)	center(Y_2)	center(Y_3)
center(Y_1)	0		
center(Y_2)		0	
center(Y_3)			0

Table 1: Point coordinates for the two-dimensional dataset used in Tasks 2 and 3.

ID	p_1	p_2	p_3	p_4	p_5	p_6
x	1	2	2	3	3	4
y	4	5	4	3	4	1

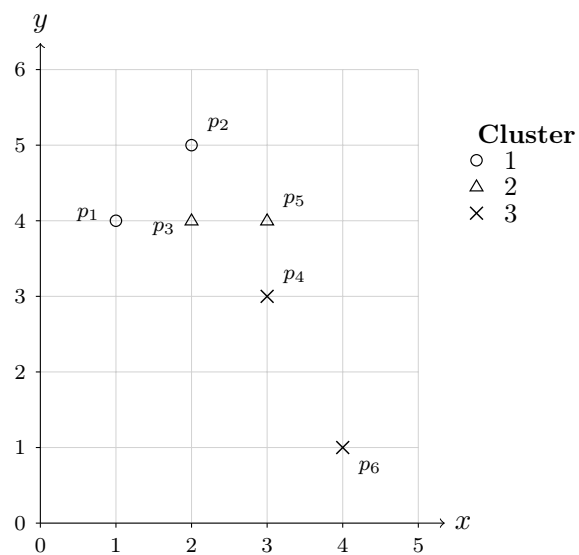


Figure 1: Clustering result for the given two-dimensional dataset used in Tasks 2 and 3.